AISI 1060 Steel, as rolled

Categories: Metal; Ferrous Metal; Carbon Steel; AISI 1000 Series Steel; High Carbon Steel

Key Words:	DIN 1.0601, AFNOR CC 55, UNI C 60 (UK), B. S. 060, MIL SPEC MIL-S-16974, SAE J403, SAE J412, SAE J414, UNS G10600, AMS 7240, ASTM A29, ASTM A510, ASTM A576, ASTM A682
Vendors:	No vendors are listed for this material. Please click here if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	7.85 g/cc	0.284 lb/in³	

Mechanical Properties	Metric	English	Comments
Hardness, Brinell	241	241	
Hardness, Knoop	265	265	Converted from Brinell
Hardness, Rockwell B	97	97	Converted from Brinell
Hardness, Rockwell C	22	22	Converted from Brinell
Hardness, Vickers	254	254	Converted from Brinell
Tensile Strength, Ultimate	814 MPa	118000 psi	
Tensile Strength, Yield	485 MPa	70300 psi	
Elongation at Break	17 %	17 %	In 50 mm
Reduction of Area	34 %	34 %	
Modulus of Elasticity	205 GPa	29700 ksi	Typical for steel
Bulk Modulus	160 GPa	23200 ksi	Typical for steel
Poissons Ratio	0.29	0.29	Typical For Steel
Shear Modulus	80.0 GPa	11600 ksi	Typical for steel
Izod Impact	18.0 J	13.3 ft-lb	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000180 ohm-cm @Temperature 20.0 °C	0.0000180 ohm-cm @Temperature 68.0 °F	condition of specimen unknown

Thermal Properties	Metric	English	Comments
CTE, linear	11.0 µm/m-°C @Temperature 20.0 - 100 °C	6.11 µin/in-°F @Temperature 68.0 - 212 °F	
	12.2 µm/m-°C @Temperature 0.000 - 300 °C	6.78 µin/in-°F @Temperature 32.0 - 572 °F	
	13.7 µm/m-°C @Temperature 0.000 - 500 °C	7.61 µin/in-°F @Temperature 32.0 - 932 °F	
Specific Heat Capacity	0.502 J/g-°C @Temperature >=100 °C	0.120 BTU/lb-°F @Temperature >=212 °F	condition unknown
	0.544 J/g-°C @Temperature 150 - 200 °C	0.130 BTU/lb-°F @Temperature 302 - 392 °F	
Thermal Conductivity	49.8 W/m-K	346 BTU-in/hr-ft²-°F	Typical steel

Component Elements Properties	**Metric**	**English**	**Comments**
Carbon, C	0.55 - 0.66 %	0.55 - 0.66 %	
Iron, Fe	98.35 - 98.85 %	98.35 - 98.85 %	As remainder
Manganese, Mn	0.60 - 0.90 %	0.60 - 0.90 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Sulfur, S	<= 0.050 %	<= 0.050 %	

References for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's terms of use regarding this information. Click here to view all the property values for this datasheet as they were originally entered into MatWeb.